

Project Plan

Project Overview

The goal of this project is to build smart trading algorithms that buy and sell 4 big stocks (AMZN, GOOG, INTC, MSFT), trading **exactly 1 share per minute** for each stock. The buy algorithm aims to minimize the average purchase price, and the sell algorithm aims to maximize the average sale price.

The standard TWAP benchmark also trades exactly 1 share per minute, executing each trade **exactly at the beginning of each minute** without adapting to market conditions.

We use high-frequency limit order book data, which provides the top 5 price levels and corresponding sizes along with order updates, to extract short-term market signals such as bid-ask spreads, order imbalances, changes in depth, rapid price movements, and recent trading activity.

Unlike TWAP, our algorithms can choose the **best timing within each minute** based on the current market. Strategies can be stock-specific, cross-stock, adaptive, or hybrid, combining different signals and models.

Performance is tested on separate out-of-sample data, looking at average buy/sell prices, improvement over TWAP, and results per stock and overall. The main aim is to see if using order book info can consistently beat TWAP.

Milestone 1: Idea Generation → Rapid Testing (35%) (April 20 - April 21)

- 12-minute **Zoom** presentation + 3-minute Q&A
- 2-page bullet-point / plots summary

Goal: Generate 5-6 high-level ideas / signals, rapidly test them, and refine 1–3 promising strategies

Objective

- Start broad with 5-6 **high-level execution strategy ideas / signals**
 - Quickly test, refine and filter to the strongest 1-3 and present preliminary performance
 - Focus first on **breadth and creativity**, then on **depth and promising strategies**.
 - No finalized algorithm implementation needed yet – goal is exploration, learning, and prioritization.
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Phase 1: Idea Generation (Top-Down)

- Resources: SSRN, arXiv, Quantpedia, GitHub, Reddit, intuition, creativity
 - Think about the big picture, break ideas into concrete components
 - For each idea, decide on core intuition, potential features/signals, high-level model types
 - Iteratively refine ideas ($v1 \rightarrow v2 \rightarrow \dots$).
 - Explore breadth; related ideas can be grouped to be studied together
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Phase 2: Rapid Testing and Development

- Quickly implement your ideas from Phase 1

- Construct key features/signals; test different models
 - Conduct preliminary backtests and compare against TWAP benchmark
 - Keep strongest 1–3 ideas; drop weak ideas
 - Refine strong ideas: improve features/signals, model considerations, execution logic, and adjust key parameters
 - Present preliminary performance with clean tables and well-labelled plots.
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Zoom Presentation Guidelines

- **Presentation:** 12-min talk + 3-min Q&A
 - Present 5-6 initial ideas very briefly and then focus on 1-3 strong ideas
 - Clearly summarize core intuition, features/signals, model type & execution logic, and show preliminary performance plots
 - **Report:** 2-page (max) bullet-point summary with plots (static compact version of slides)
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Presentation Grading

- **Breadth (10 pts):** Present 5–6 high-level ideas briefly, covering different angles
- **Clarity (10 pts):** High-level ideas, features/signals, models and execution logic explained clearly
- **Depth (5 pts):** Strong ideas and improvements
- **Preliminary Performance (5 pts):** Show preliminary results with clean tables and well-labeled plots and TWAP comparison
- **Presentation Interaction (5 pts):** Engaging, organized, responsive to questions and feedback

Final Milestone: Finalized Strategy and Performance (65%) (May 12 - May 15)

- 18-minute **in-person** presentation + 4-minute Q&A
- 2-page bullet-point / plots / table summary

Goal: Finalize smart execution algorithms, test them rigorously, and clearly show their out-of-sample performance, in comparison with TWAP benchmark

Objective

- Explore your 1-3 smart ideas from Milestone 1 more deeply
 - Combine your ideas into the final execution strategy logic
 - Refine, modify, fine-tune, and optimize your strategy; and finalize your implementation (clear rules, parameters, running code)
 - Clearly present a performance plots and table that shows buy improvement / sell improvement / overall improvement for each stock, both in-sample and out-of-sample
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Phase 1: Deeper Exploration of Strong Ideas

- Revisit the core intuition for each strong idea
 - Improve the design - you can design more features and signals - and try to improve your models
 - Refine and adjust the execution logic
 - Test the improved versions and evaluate preliminary performance
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Phase 2: Combining Strong Ideas into One Final Algorithm

- Study the performance of strong ideas and how they relate to each other
 - Think about how to combine your strong ideas – some practical ways are
 - **Pick the best idea:** a performance based selection, you can choose the most stable one that performs most reliably out-of-sample and discard the others
 - can be useful if one strategy is clearly stronger and dominates others
 - **Signal level combination / allocation:** take useful signals / features from strong ideas and combine them into a single decision system – for example; by a weighted score, confidence-weighted allocation, linear / nonlinear model or voting based model that considers all signals together
 - can help your algorithm with robustness; especially if your strong ideas are complementary and capture different aspects of microstructure
 - usually the most practical approach
 - **Regime-based switching:** identify market regimes and use different strategies in different market regimes – you can build a higher-level meta model / rule that decides which strategy to activate based on market state, features, or recent performance
 - can adapt well to adaptive market environments, yet performance can vary
 - **Hierarchical / layered decision logic:** You can layer ideas in a sequence, refining the timing, also acting as a filter or safety check
 - creates a clear if-then tree narrowing down when to execute
 - **Stock-specific allocation:** use different ideas to different stocks based on which idea performs best for that stock's behavior, instead of using the same strategy for all
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Optimization and Implementation

- Refine, modify, fine-tune, and optimize your strategy; and finalize your implementation (clear rules, parameters, running code)
- You can do robustness checks and reflect on where/why your strategy does well and where/why it struggles - and whether results are stable or fragile...

- Have a clear, unambiguous and systematic final strategy and implementation
 - Make sure your code is clean, organized, readable and end-to-end automated
 - stable feature computation (handling edge cases, extreme values) without breaking
 - no data leakage: only use point-in-time data
 - have clear train / validation split
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Algorithm Evaluation and Out-of-sample Performance

- We will share out-of-sample data 1-2 days before final presentations
 - For each stock, evaluate performance metrics: buy improvement, sell improvement, and overall improvement – both for in-sample and out-of-sample
 - Produce clean well-labeled plots / tables
 - Reflect on and conclude your project
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In-person Presentation Guidelines

- **Presentation:** 18-min talk + 4-min Q&A
 - Recap your 1-3 strong ideas from Milestone 1 and present any improvements
 - Present your final algorithm: clearly summarize core intuition, features/signals, models used, optimization logic, how signals are combined, and final execution logic
 - Present performance plots and table showing both in-sample and out-of-sample performance metrics for each stock
 - Conclude with reflection and key takeaways
 - **Report:** 2-page (max) bullet-point summary with plots (static compact version of slides)
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Presentation Grading

- **Depth (10 pts):** Strong ideas are presented clearly with improvements

- **Final Strategy Design (20 pts):** Final algorithm is presented clearly, with intuition, features/signals, models used, optimization logic, and how signals are combined
 - **Final Performance (15 pts):** Show in-sample and out-of-sample performance results with clean tables and well-labeled plots and TWAP comparison
 - **Clarity and Organization (10 pts):** Story and logic is easy to follow, slides are well-organized
 - **Presentation Interaction (10 pts):** Engaging, responsive to questions and feedback
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